

Mind Over Matter: Using Artificial Intelligence to Predict the Shape of Proteins

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1 Introduction

Stuyvesant requires that of its students who want the Stuyvesant Diploma, i.e. the very small sticker on our regular high school diplomas, to take at least one semester of Computer Science. For those who want to major in biology, the class may seem like a waste of time. But before you run off to your friends and complain about how much you hate the class, consider the possibility that Computer Science can hold the key to future advancement in medicine. Artificial intelligence (AI) was able to predict the structure of proteins' 3D shape better than any other prediction mechanism that exists to this day.

2 A Brief History

Since the 1960s, researchers have been scratching their heads to try and find methods to predict a protein's 3D shape. At first they thought that if they could work out all the individual interactions within a protein sequence, they could predict the 3D shape. As researchers know now, there are hundreds of amino acid sequences per protein and numerous ways each pair of amino acids interact, making the number of possible structures per sequence astronomical. Many researchers tackled this problem, but progress was slow.

3 Current Methods and Their Limitations

Some experimental techniques such as x-ray crystallography and cryo-electron microscopy (cryo-EM) have emerged to help predict the 3D structures of proteins. X-ray crystallography works by obtaining a three-dimensional molecular structure from a crystal. A purified sample of a protein, for instance, is crystalized, and then an x-ray beam would shine through it. The diffraction patterns that result can later be used to analyze the structures that make up the crystal and in turn, the shape of the protein. Cryo-EM is a special microscope that can see the shape of complex proteins at molecular resolution of a frozen hydrated specimen. These methods take up to months or years and don't always work. As of right now, we know the shape of only about 170,000 of the more than 200 million proteins discovered across life forms.

4 Critical Assessment of protein Structure Prediction (CASP) Competition

In 1994, John Moult, and his colleagues started CASP, which takes place every two years, to find potential new ways to predict protein structure. Entrants are given amino acid sequences for about 100 proteins whose structures are not known. For each sequence, certain groups compute a structure, while others determine it experimentally. The organizers then equate the computational predictions to the lab results and assign a global distance test (GDT) score to the predictions. Scores of 90 or higher on a scale of 0 to 100 are considered comparable to experimental methods. When the competition first launched, many groups were able to predict the structures for small, simple proteins. For larger, more challenging proteins, however, GDT scores were as low as 20.

5 DeepMind's AlphaFold

DeepMind first competed in CASP in 2018 with its algorithm called AlphaFold. AlphaFold used a comparative strategy. AlphaFold also incorporated a computational approach called deep learning in which the software is trained on vast data troves to learn patterns. In this case, the vast troves were sequence and structure proteins. However, at that point the predictions were still too coarse to be helpful. To improve, they combined deep learning with an attention algorithm that would mimic the way a person would solve a jigsaw puzzle: first connecting pieces in small clumps then searching for ways to join the clumps into a larger whole. In this case, amino acids were first searched in large clumps then they searched more ways into joining the clumps into a larger whole. Working with a computer network built around 128 learning processors, they were able to train the algorithm to know all 170,000 protein structures. In the CASP contest, DeepMind received a GDT median score of 92.4, higher than any other competitors ever before, paving a way for more innovation.

6 More on how AlphaFold Works

AlphaFold was designed to use neural networks to predict physical properties of a protein from its genetic sequence. The properties which DeepMind's networks predicted were the distance between amino acids and the angles between chemical bonds that connect the amino acids. DeepMind

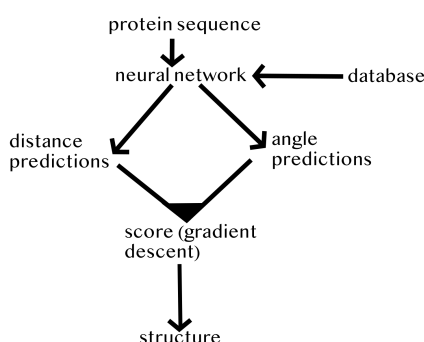
trained a network to predict the distribution of distances between every pair of residues in a protein. The probabilities were then combined into a score that estimates how accurate a proposed protein structure was. DeepMind also trained another neural network that uses all distances in aggregate to estimate how close the proposed structure is to the right answer.

Using the score functions, DeepMind researchers were able to search the protein landscape to find structures that match their predictions. Since DeepMind built their first method on techniques commonly used in structural biology, they trained a neural network to invent new fragments, which

were continually used to improve the score of the proposed protein structure.

The second method optimized scores through a gradient descent. Gradient descent is a mathematical technique commonly used in machine learning for making small, incremental improvements. (Gradient descent is based on the observation that if the multivariable function $F(x)$ is defined and differentiable in the neighborhood of point a , then $F(x)$ decreases fastest if one goes from a in the direction of the negative gradient of F at a , $-\nabla F(a)$.) This technique was later applied to entire protein chains rather than to pieces that must be folded separately before being assembled into a larger structure, to simplify the prediction process.

This entire process can be visualized by this flow chart:



The AlphaFold version used at CASP13 is available here:

https://github.com/deepmind/deepmind-research/tree/master/alphafold_casp13 for anyone interested in learning more.

7 The Application to Medical Advancement

As you may know, proteins are large complex molecules that are essential to everyday life. Nearly every function our body performs relies on proteins and how much they move or change. Each protein's shape determines what its "job" is inside the human body. Since we are in a pandemic, let's use antibodies as an example. Antibody proteins are Y-shaped and form hooks. Because of these hooks, antibodies are able to latch on to viruses and bacteria in order to detect and tag them. Other proteins, like collagen protein, are shaped like cords. Their shape allows them to transmit tension between cartilage, ligaments, and bones which helps hold the body together and enable it to function properly. There are tens of thousands of proteins in the human body alone, each serving a specific purpose, and the recipe to make all those proteins is found in genes.

DNA uses genes in order to code for and make proteins. And just as a missed parentheses while writing code can be catastrophic to the program, an error in the genetic recipe can lead to the

malformation of proteins, which can lead to disease or death of an organism.

For instance, cystic fibrosis is a genetic disorder caused by a gene mutation that often removes the amino acid phenylalanine in the 508th position of the cystic fibrosis transmembrane conductance protein. This one deletion of an amino acid causes an individual with this disease to have a faulty channel protein. Without a working channel protein, the body's salt and water concentration gets thrown off, leading to sticky mucus that clogs airways. The expected life span of a person with cystic fibrosis is about 44 years old which is extremely short compared to the norm.

Although scientists know which genes make which proteins, they don't know the shape of the protein and how exactly it folds. If scientists found a way to predict a protein's specific shape, drugs could be created to match a protein's specific shape and help treat diseases like cystic fibrosis.

There is little or no way scientists can predict the shapes of every single protein in an efficient amount of time which is why precision medicine has been so limited. However, programs like DeepMind's AI algorithm, AlphaFold, completely change the game. With the use of previous genomic data and the use of neural networks, AlphaFold was able to predict the shape of complex proteins better than any previous technique, accurately and efficiently. Knowing protein shape can help scientists and researchers create more targeted drugs and therapies and treat disease and fatal conditions caused by a malformation of a protein, saving and improving human lives.

8 Final Thoughts

After researching for hours for theoretical ways to cure genetic disease as a part of my genetics class, I never considered predicting a protein's shape and creating a drug tailored to that shape. I thought cutting out the faulty DNA and replacing it with a new one would be the way to go. Even though I knew that cutting out DNA was risky, that was the only way I could think of to "cure" a disease. I thought that researchers could always just create more precision medicine tailored to a person's individual genetics if cutting and replacing DNA were not an option. Creating drugs to help a protein specific shape is theoretically safer since the person's original DNA isn't tampered with. Without knowing the protein's shape, scientists have spent too much time trying to fix the root of a faulty protein rather than just treating the protein itself. With these new AI programs, computer science can fundamentally change the future of precision medicine and its potential applications.

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